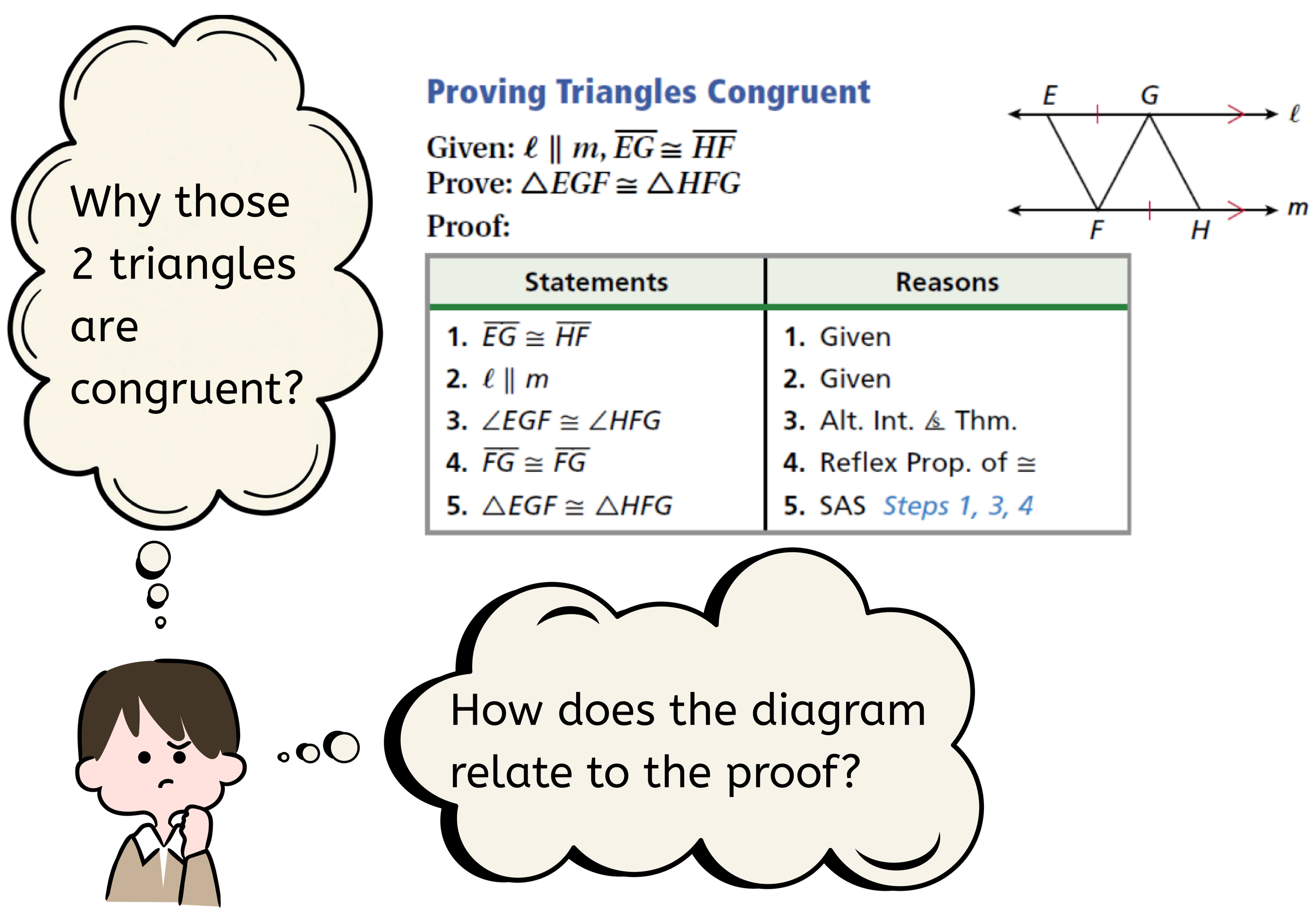
AUTOMATING TEXTBOOK PROOF CONVERSION AND DATASET GENERATION FOR GEOMETRY LEARNING

Quyen Tran[1], Hwei-Shin Harriman[2], Dominik Moritz[2], Joshua Sunshine[2]

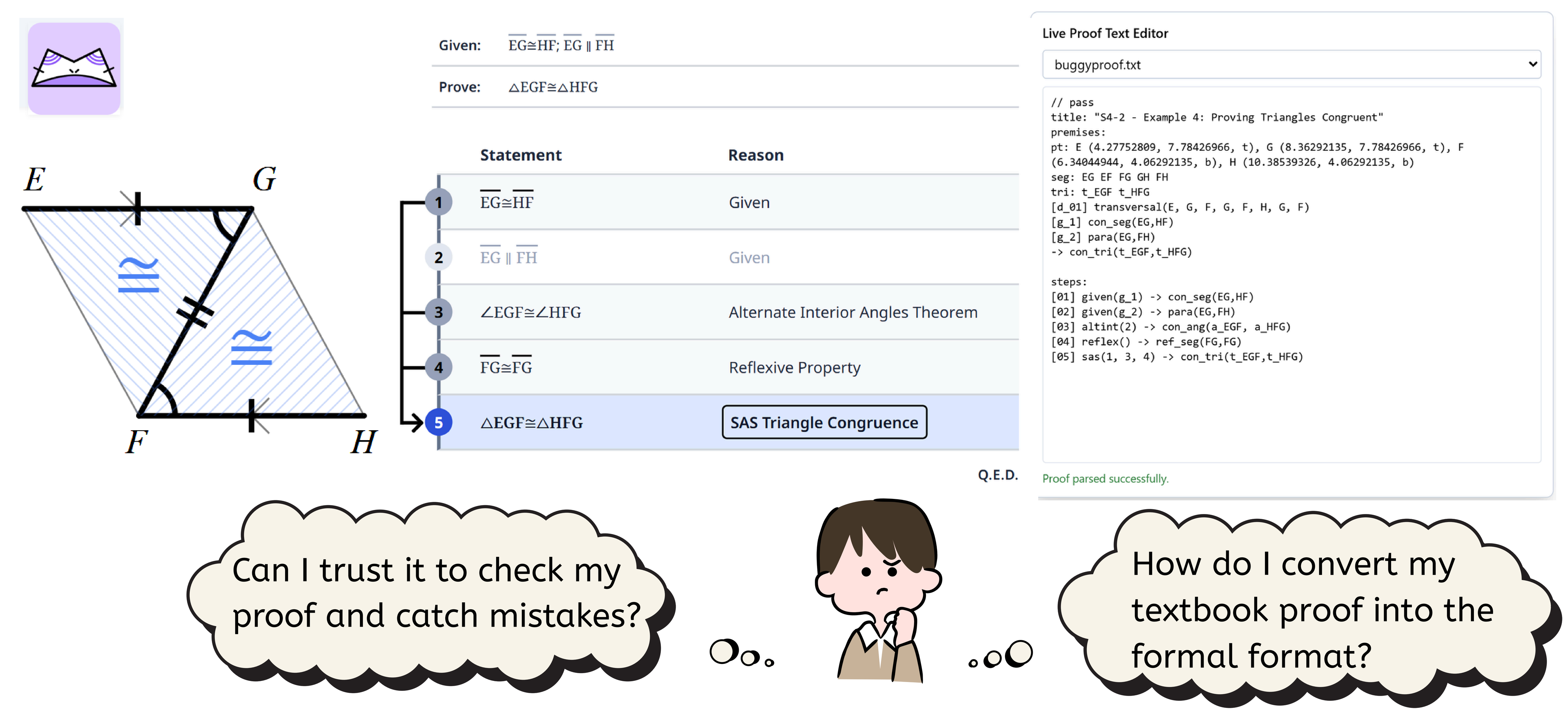
[1] DePauw University [2] Carnegie Mellon University

Motivation

Say this student is looking at this high school geometry textbook problem:



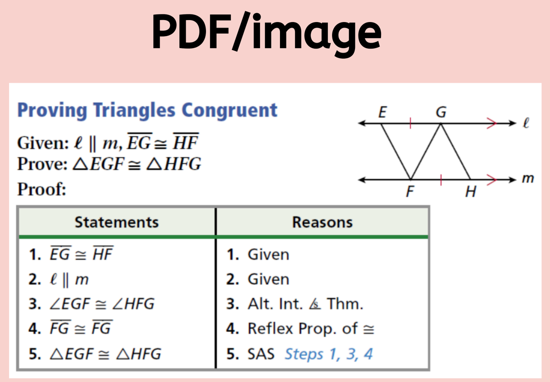
Our tool, ENDER [1], can help students connect diagrams, statements, and reasons, but requires proofs in a specific formal syntax.



Extraction Pipeline

Proof text extraction

PDF/image



Checker-guided prompt
Context: {reasons+definitions + examples}.
Rule: Use only supported syntax.

→ **LLM**
Prompt: Identify geometry proof tasks and convert them into checker-valid proofs

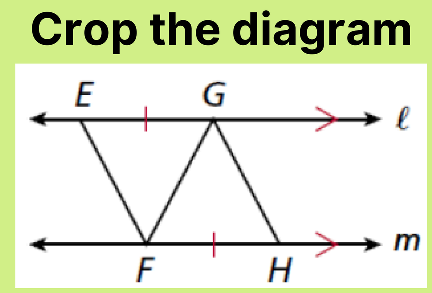
→ **Ender proof text + point labels**

```
premises:
pt: E (0, 0, tl), G (0, 0, tr), F (0, 0, bl),
H (0, 0, br)
seg: EG GH HF FG EF
tri: t_EGF t_HFG
[d_01] transversal(E, G, G, G, F, H, F, F)
[g_1] para(EG, FH)
[g_2] con_seg(EG, HF)
-> con_tri(t_EGF, t_HFG)
steps:
[01] given(g_2) -> con_seg(EG, HF)
[02] given(g_1) -> para(EG, FH)
[03] altint(2) -> con_ang(a_EGF, a_HFG)
[04] reflex() -> ref_seg(FG, FG)
[05] sas(1, 3, 4) -> con_tri(t_EGF, t_HFG)
```

Diagram extraction

LLM
Prompt: Match each proof to its diagram and return a bounding box containing the complete diagram.

→ **Crop the diagram**



→ **Point-coordinate extraction [2]**

```
pt:
E (4.27752809, 7.78426966, t),
G (8.36292135, 7.78426966, t),
F (6.34044944, 4.06292135, b),
H (10.38539326, 4.06292135, b)
```

→ **Complete Ender proof**

```
premises:
pt: E (4.27752809, 7.78426966, t), G (8.36292135,
7.78426966, t), F (6.34044944, 4.06292135, b), H
(10.38539326, 4.06292135, b)
seg: EG EF FG GH FH
tri: t_EGF t_HFG
[d_01] transversal(E, G, F, G, F, H, G, F)
[g_1] con_seg(EG,HF)
[g_2] para(EG,FH)
-> con_tri(t_EGF,t_HFG)

steps:
[01] given(g_1) -> con_seg(EG,HF)
[02] given(g_2) -> para(EG,FH)
[03] altint(2) -> con_ang(a_EGF, a_HFG)
[04] reflex() -> ref_seg(FG,FG)
[05] sas(1, 3, 4) -> con_tri(t_EGF,t_HFG)
```

Generate incorrect proofs for checker evaluation

Correct Proof (Ender's format) → **Construct mutation structures (reason graph and predicate trees)** → **Define mutation methods and constraints** → **Apply mutation types** → **Incorrect proofs**

Mutation types

1. Change a statement or geometric object

Before: con_seg(AB, CD)

After: sim_seg(AB, CD)

2. Rearrange point labels

Before: con_ang(a_ABC, a_DEF)

After: con_ang(a_ACB, a_DEF)

3. Remove a necessary proof step before the conclusion

Before:	After:
[01] given(g_1) -> midpt(B,AC)	~~[01] given(g_1) -> midpt(B,AC)~~
[02]	[01] .....

Results

81 textbook proofs converted

696 wrong proofs created for evaluation

726T wrong proofs can be created

11 proof-checker bugs identified

Limitations

Current extraction relies on costly LLM calls; some proofs need manual correction, and the dataset is limited to selected chapters of Holt Geometry and synthetic errors.

Future work

Ongoing work will broaden extraction and mutation methods, compare synthetic errors with student work, and support uploaded proof images for automatic interactive feedback.

[1] Harriman, Hwei-Shin, et al. "Ender: Scaffolding Geometric Proof Comprehension with Automated Visualization of State"

[2] Lee, Sally. "Automating Geometric Proof Feedback"

ENDER Github

Sally's poster